Primitive residue extension for a fixed finite acting group

Private L100 applicability component

This note specializes the previously accepted Topic D arguments: anchored Helly, panel recognition, subgroup transport and maximal augmentation. The printed augmentation theorem there imposes local two-transitivity. Here that extra restriction is removed explicitly; primitivity alone is used. The exact general double-coset test already present in the phase-six and phase-seven programs is retained. No new enumeration of the $\mathrm{PSL}(3,4)$ geometries is asserted in this note.

All cosets below are right cosets, with incidence defined by nonempty intersection. Inverting every group element converts the source's left-coset convention to this convention and preserves incidence. Geometries are firm when each panel has at least two objects, residually connected, and flag-transitive under the specified acting group. RWPRI means that in each positive-rank residue the specified residue stabilizer acts primitively on at least one type. The rank-two axiom $(\mathrm{IP})_2$ means that each rank-two residue is a partial linear space or a generalized digon. A local action may have a kernel; subgroup maximality still characterizes its primitivity.

Theorem 1 (Broad primitive panel recognition)**.** *Fix a finite group G, $B \le G$ and distinct subgroups $P_i > B$ indexed by a finite set I. Put $K_\varnothing = B$ and $K_S = \langle B, P_i : i \in S\rangle$. The family defines a firm, flag-transitive, residually connected RWPRI $(\mathrm{IP})_2$ geometry for its acting group K_I, with chamber stabilizer B, precisely when:*

1. *$K_S \cap K_T = K_{S\cap T}$ for all $S, T \subseteq I$;*
2. *for every nonempty S, some $K_{S\setminus\{i\}}$ is maximal proper in K_S;*
3. *for every $|S| \ge 3$, the deletion parabolics $(K_{S\setminus\{i\}})_{i\in S}$ have typed coset Helly;*
4. *for every two-element S, the corresponding coset graph is a partial linear space or a generalized digon.*

Conversely every geometry with these properties arises from its actual panel stabilizers in this way.

Proof. The first condition, together with proper panels and the top Helly condition, is the accepted panel recognition theorem. It identifies the stabilizer of the complementary base flag with K_S, and identifies each residue with the corresponding coset geometry inside K_S. The singleton cases of the second condition give proper maximal inclusions $B < P_i$ and hence firmness. All other cases of that condition give exactly RWPRI, by the block–subgroup correspondence for a transitive coset action; its kernel does not affect that correspondence. The fourth condition gives exactly $(\mathrm{IP})_2$ in the identified residues.

For necessity, use actual flag stabilizers. Their intersections give the first condition. The stabilizer of a residue is generated by its panel stabilizers: in rank two this is the coset-path connectedness argument; in higher rank, connectedness and induction first give generation by the maximal parabolics, then by their panel stabilizers. Equivalently one can lift an incidence path to adjacent chambers, using the already connected lower-rank residues. This retains the common chamber stabilizer B at every step. Flag transitivity gives typed Helly, and every residue inherits all four conditions. Thus no local two-transitivity premise is used. □

Lemma 1 (The general rank-two test)**.** *Let $B = P \cap Q$ and $K = \langle P, Q \rangle$. The coset graph is a digon if and only if $|P||Q| = |K||B|$. Otherwise it is a partial linear space if and only if, for every nontrivial double coset BxB of B in Q,*

$$|\{Qh : h \in P\} \cap \{Qhx : h \in P\}| = 1.$$

Proof. The neighbours of P are the Qh with $h \in P$; their number is $[P : B]$. This equals the entire number $[K : Q]$ of opposite objects precisely in the displayed digon case. Otherwise a quadrangle exists precisely when two distinct P-type objects have two common neighbours. Translate one common neighbour to Q, and then translate one of the two objects to P by an element of Q. The other is Px for $x \in Q \setminus B$. The remaining stabilizer of the base incident pair is B, and its orbits on these objects are the double cosets $B\backslash Q/B$. The two neighbour sets are the displayed sets. They already share Q, so the required bound of at most one is exactly equality with one. This treats every double coset, without a two-transitivity shortcut. □

Theorem 2 (Complete primitive augmentation)**.** *Assume representatives of every G-conjugacy class of subgroups are supplied, together with representatives of every K-conjugacy class of maximal subgroups for each representative K. For each Borel representative B, let $\mathcal{P}(B)$ contain* all *actual overgroups P in G with B maximal proper in P. Apply the phase-seven augmentation with* `REQUIRE_2T=false`*, its general rank-two test, and all four recognition conditions. If every Borel run reaches its certified empty next level, the outputs with join G are exactly the requested geometries, represented as subgroup families, modulo inner conjugacy and permutation of types.*

Proof. UD.Transport enumerates all maximal overgroups of any given subgroup from the retained inclusion data. There is no local-action filter in $\mathcal{P}(B)$. In a target of rank r, RWPRI at the top supplies a maximal deletion H in the target join K. Its rank-$(r-1)$ panel family is a residue satisfying the same conditions, so induction reaches it. Its missing panel P lies in K but not in H; since H is maximal, adjoining P has join exactly K. Every such maximal overgroup and every such panel is tried. The pointwise stabilizer of the old tuple in the image of $N_G(B)$ acts on these extensions; taking its orbits loses no family. Using this possibly smaller stabilizer instead of the setwise stabilizer only leaves duplicates. The program's final successive-transporter test and type permutations remove exactly those duplicates. Its order and pair-join signatures are necessary invariants, never sufficient equivalence tests. Conversely each accepted family passes the exact recognition conditions.

The intersection condition forces every new panel to enlarge the join: $P_i \leq K_S$ for $i \notin S$ would imply $P_i = P_i \cap K_S = B$. Consequently the rank is at most $\Omega([G : B])$, counting prime factors with multiplicity. The extra next level must therefore be empty. A timeout, missing Borel, unfinished anchor, or numeric cap before this terminal test is not an exhaustion certificate. Finally every Borel is conjugate to an input representative, and conjugating a family also conjugates its Borel. □

Exact implementation boundary. All seven top-level function definitions, from `IsMaxPair` through `RunBorel`, and the maximal-inclusion initializer are extracted byte-for-byte from `phase7_sporadic_search.g`; the legacy file is unchanged. The only new interface supplies the complete explicit group and subgroup representatives and fixes the broad-mode flag. `FlagTransitive` uses the anchored Helly test: for each distinguished parabolic it intersects the sets of cosets meeting the other base parabolics and compares their cardinality with the contained set of cosets meeting their intersection. All subfamilies of size at least three are checked. `ValidPanels` checks every subgroup intersection and every maximal deletion. `RankTwoAdmissible` additionally checks the two-deletion RWPRI condition. `IsMaxPair` is called only for known inclusions. Prime index is an exact sufficient shortcut; all other branches use maximal-subgroup transport or the actual primitive coset action. The union of maximal subgroup orders is only a necessary pruning test.

L100 retained inputs and equivalence. The installed TomLib1.2.11 table named `L3(4)` has95 subgroup classes. Its underlying degree21 permutation group has order20160 but is not literally the retained named action. A separately checked homomorphism given by images of its two generators is bijective onto the retained group. Transporting the complete representative and inclusion lists by that map preserves all conditions. TomLib's complete subgroup classification and GAP's exact maximal-subgroup, transporter and group algorithms remain named retained premises, not conclusions of the small controls. No claim is based merely on equal orders or names.

The first census would produce inner-conjugacy/type classes. A later exact $\mathrm{Aut}(G)$ fusion would classify the specified-action objects up to automorphism of G and type permutation. This alone is not asserted to classify bare incidence geometries: an arbitrary incidence isomorphism need not normalize the specified subgroup G. An additional faithful incidence-isomorphism comparison or a proved uniqueness-of-G argument would be required for that assertion. Full automorphism and correlation groups, displayed kernel/group names, all remaining constructions, and the whole source contract are separate obligations.

Bounded controls and attribution. Literal multiplication-table controls independently enumerate all subgroups and all panel subsets in $S_3, C_2^2, D_8, A_4, C_3, C_3^2, C_2^3$. They compare every partial and full family, all Borel representatives, terminal empty levels, and normalizer/type orbits with the unchanged engine. They use direct coset incidence for Helly and $(\mathrm{IP})_2$, not the extension recurrence. The cyclic-three controls exhibit primitive non-two-transitive panels; the elementary-abelian order-eight control includes rank three. These controls are finite checks of the new interface, not a proof of TomLib completeness or a target census. The common recognition, transporter and extension arguments are attributed to the accepted Topic D work and its classical coset-action antecedents. The independent preliminary read of the broad specialization is source/interface scrutiny only. No novelty, measurement, formalization or source-disposition credit is claimed here.